Boxuan Wang / Imperial Design Engineering

# S00: structure design & FEA

Humanoid Development Intern. I designed supports and structural details around the S00 team assembly.

## CCU: mount, seal and align

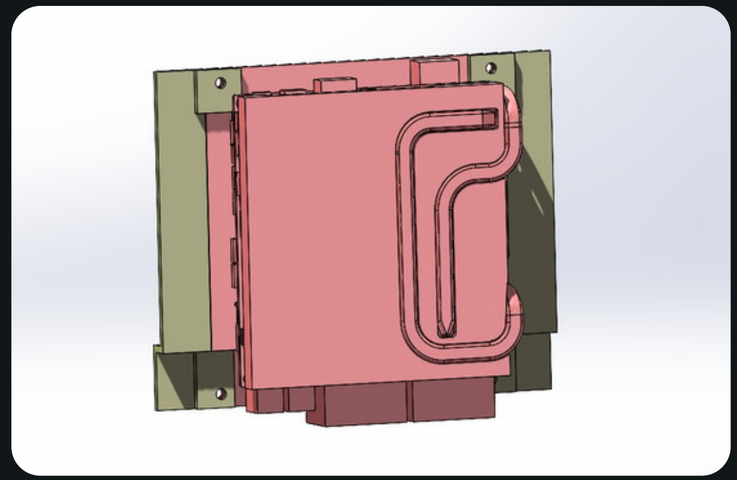
Paired identical sealing parts share M4 frame interfaces. Foam allowance and a lower locating groove support fit and assembly.

## Microphone: reinforce the span

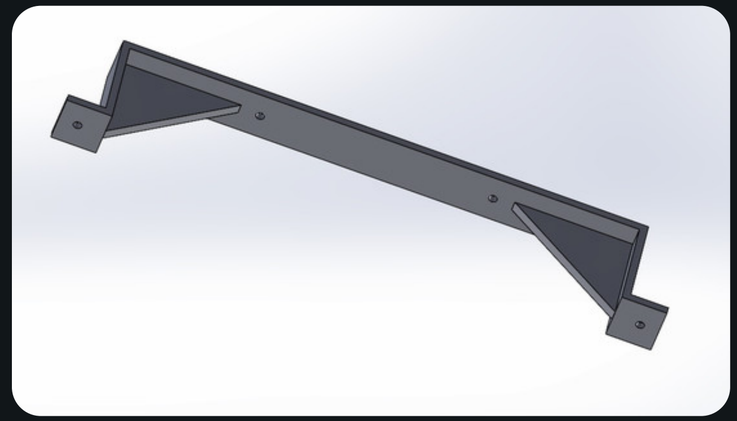
I separated the microphone support from the head-camera bracket, added corner ribs and retained central M2 fixing access.

## Keep neighbouring systems clear

Display openings and fan supports were revised around the control box. CAD decisions remain separate from physical validation.



Paired CCU sealing supports and surrounding interfaces.



Original support CAD; physical performance of this revision is unverified.



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mystify59.github.io

# Follow the main-frame load path.

I investigated load representation, the hip restraint and local shoulder mesh in the archived Ansys frame model.

## Model the assembly

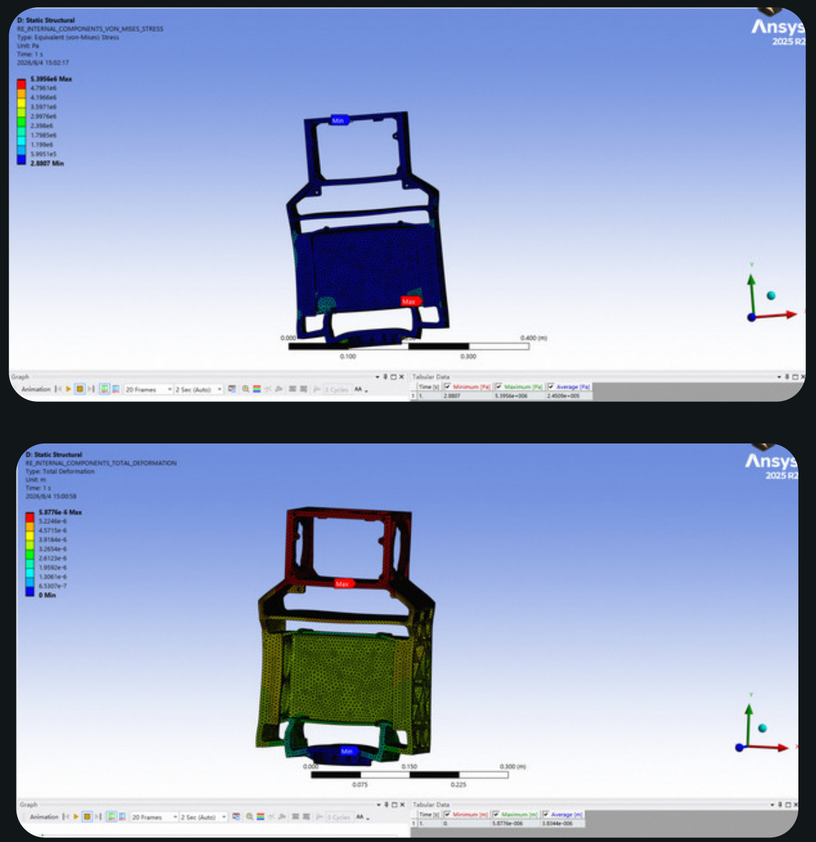
7075-T6 frame, recorded mass 4.02 kg. Gravity, two 12 kg arm masses and 5.75 kg equipment act through an idealised rigid, fixed hip.

## Retained 2 mm local mesh

## 5.396 MPa

Peak von Mises stress; maximum deformation 0.005878 mm. The governing hotspot was at the lower equipment-support transition.

2→1 mm shoulder refinement changed stress ~2.4% and displacement ~0.13%. The lower hotspot still needed refinement; no fatigue or physical validation.



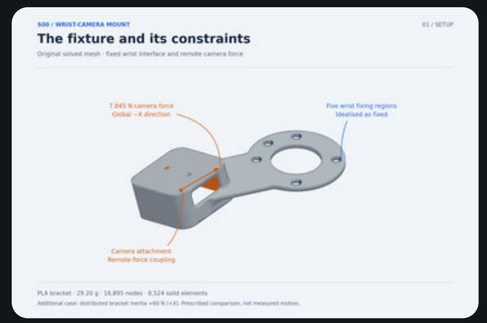

Saved 4 August results: original stress (Pa) and displacement (m) legends.

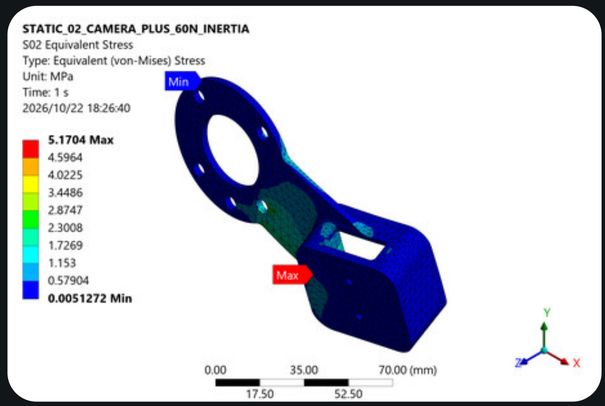
# A wrist-camera mount for VLA capture.

I designed a separate wrist fixture for a specified 0.8 kg camera and five M8 wrist locations, and participated in demonstration capture.

## Make the assumptions explicit

The 29.20 g bracket uses generic isotropic PLA, idealised fixed wrist regions and a 7.845 N camera force. The extra 60 N-equivalent bracket inertia is a specified comparison, not measured robot acceleration.

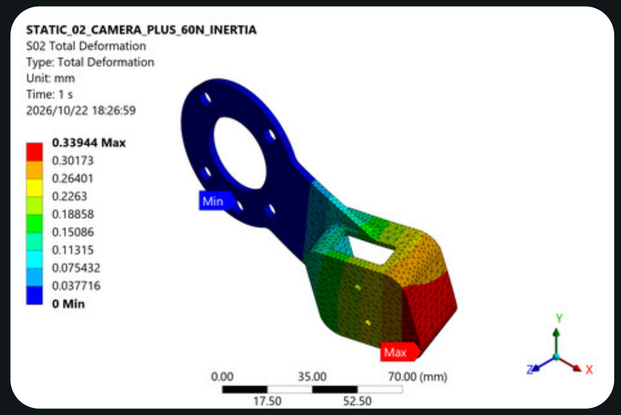



## Saved static results

Camera weight only
0.275 mm / 1.787 MPa

Weight + specified inertia
0.339 mm / 5.170 MPa




Print direction, joint flexibility, camera offset and fatigue remain unvalidated. Capture support does not establish VLA policy performance.

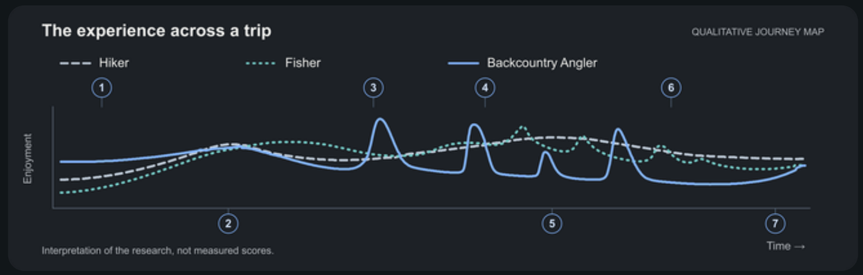
Native Ansys plots: saved 17 August results exported 22 October; no new solve. Full legends and boundary assumptions are available in the online case.

# Aquatorium: research into concepts.

Two hikers and two anglers revealed weight, access and reliability problems between walking and fishing. Our journey map synthesised these transitions.



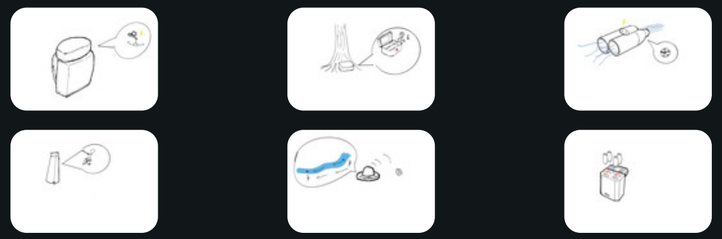
Four original participant photographs.

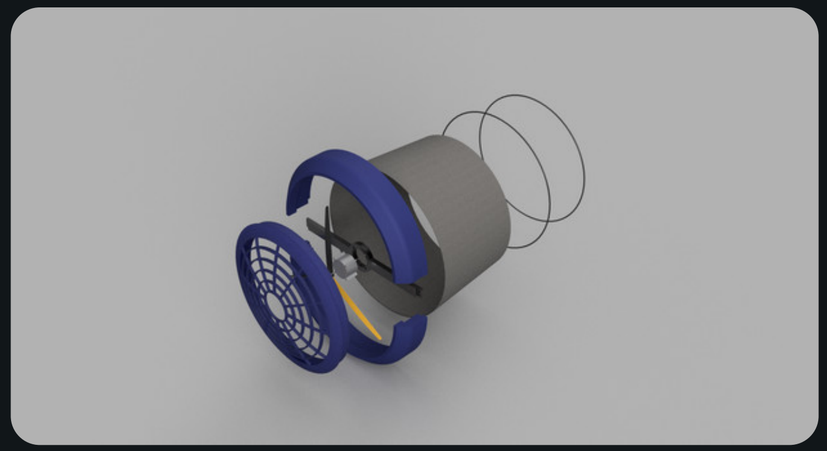



Research synthesis; curves are qualitative, not measured scores.

## My six initial concepts

I explored energy, shared supplies, combined equipment and preparation for the next stop. Feedback led me to develop a floating hydro-generator.

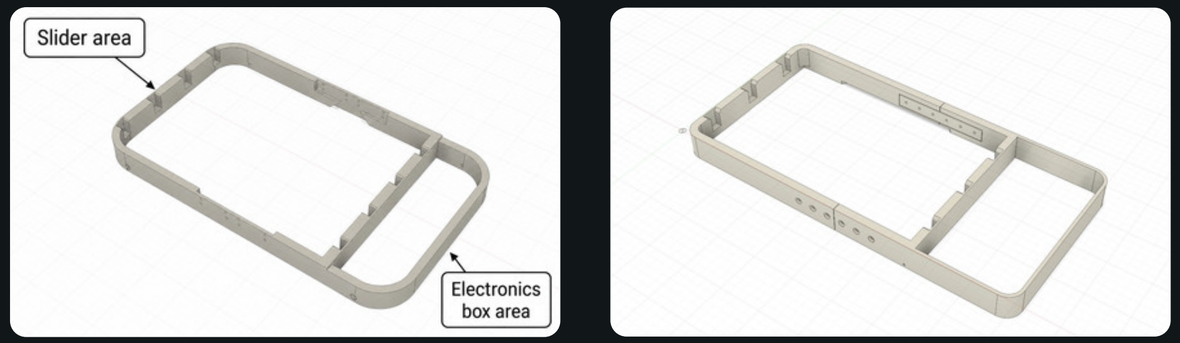




My proposed turbine CAD: removable guard, generator support and collapsible tail rings. The separate anchor held the battery. Later team prototyping found output too low for its size and weight, prompting the shift to fish transport.

# Develop the system. Plan its assembly.

The later carrier, frame and enclosure are team development. We moved from four fish-bag bays to three, freeing electronics space and improving slider access.




Earlier four-bay frame → revised three-bay arrangement.

## Three enclosure prototypes



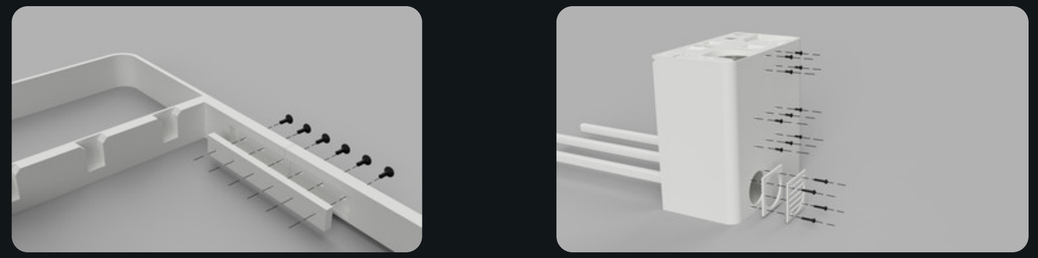
Basic box → test layout → refined openings and mounts.

## Design for manufacture

Proposed 2 mm shell, 3 mm supports and 1° draft, with radiused corners and controlled wall thickness.

## Design for disassembly

Removable M4 frame splice plates, a screw-mounted outer bag and a separate electronics module preserve access for cleaning and replacement.

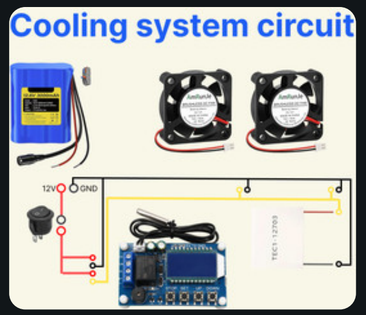


Team CAD proposals; tooling and repeated disassembly tests remain future work.

# Prototype the functions. Close the loop.

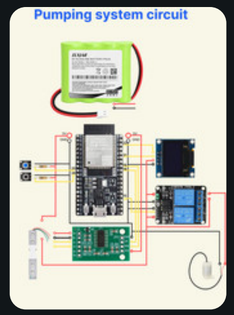
## Cooling: feedback and switching

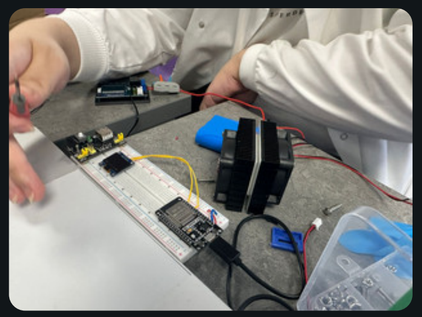
A 12 V battery supplies the XY-T01 controller, Peltier module and two fans. Set target → read probe → relay switches cooling at the reported +1°C threshold → monitor and repeat.




## Pumping: a separate control loop

An ESP32-S reads load cells through HX711 and controls the R320 pump through a relay. Buttons and display support selecting and checking the fill target.







## Load. Fill. Carry. Fold.

Separate bags, controlled filling, oxygen and cooling in a collapsible concept. Reported volume: 22 L → 8 L.

Test sensor: 25°C→20°C in ~3 min. Battery estimate: ~47 min continuously or ~4 h at assumed 20% duty; not measured endurance.

Left: physical bench prototype. Centre/right: final concept renders. Team: Joseph Wang, Joe Chen, Jason Wang and Peppe Aphiphunya.

# Street Pulse: feeling live music.

Modular floor units combine vibration and light, inviting audiences to feel a street performance beneath their feet.

## Research, design, make and test

I carried out almost all user research, designed the tessellation and exterior, prototyped the electronics, and carried out all testing and assembly.

## One functional demonstration module

Other team members developed the software. The wider tessellated installation remained a concept.

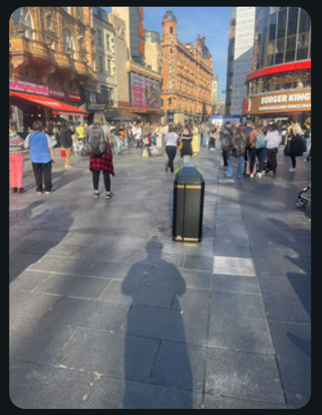


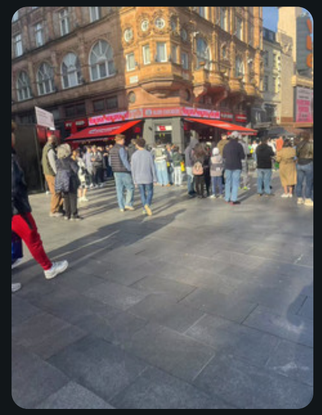
Unreal Engine street simulation: intended experience, not a built public installation. Team JJAGernaut.

# User research: from people to place.

## 01 / Interviews

We observed and interviewed performers and audiences across London. Clear views, sound quality and low-pressure participation mattered; performers needed help gathering and positioning a crowd.

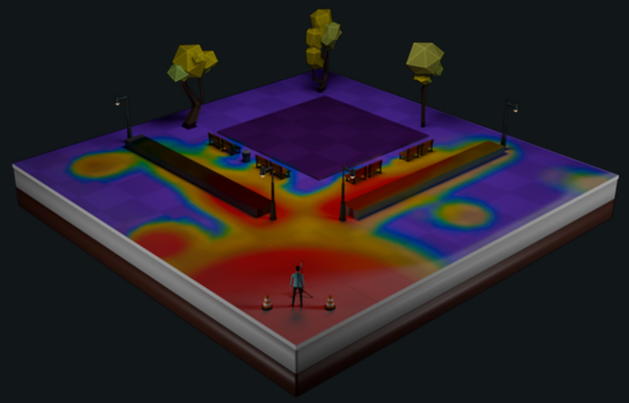


Original fieldwork: active participation and the edges of the crowd.

## 02 / Audience heat map

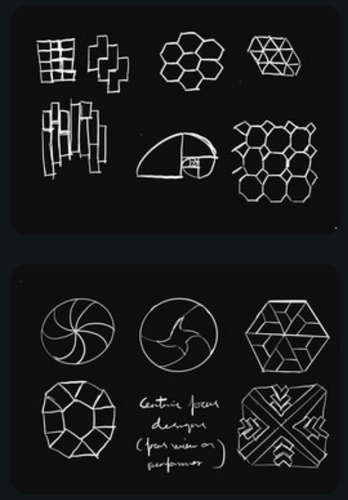
A qualitative density model brought together active participants, an outer ring and seated listeners. The layout needed to allow different levels of involvement.



Observation synthesis, not measured audience density or proof of retention.

## 03 / Tessellation studies

I explored repeatable tile geometries and exterior treatments. Co-design with 1:30 pieces informed trapezium units with perimeter lighting.

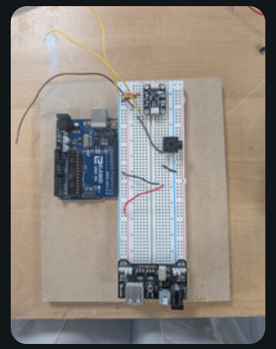


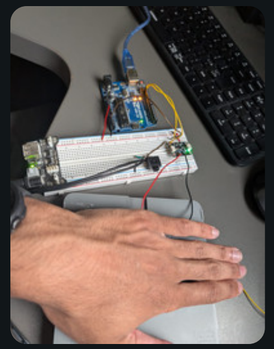
Original sketches: different spaces, one repeatable module.

# Build the sensation. Test the experience.

## The physical prototype

I prototyped electronics and assembled the module. Testing favoured 3 mm acrylic over 5 mm surfaces; firmer feet replaced soft mats that let actuators touch the floor.





## 20 people / choose the rhythm

Participants ranked three patterns with the same song. Selected rhythmic elements were preferred to metronome or full-signal vibration.

Original prototype photos. Software was developed by other team members.

## 19 people / experience it in context

Participants stood on the working vibration plate while watching a 3D street-performance video through a VR headset. The physical sensation and Unreal scene were evaluated together.

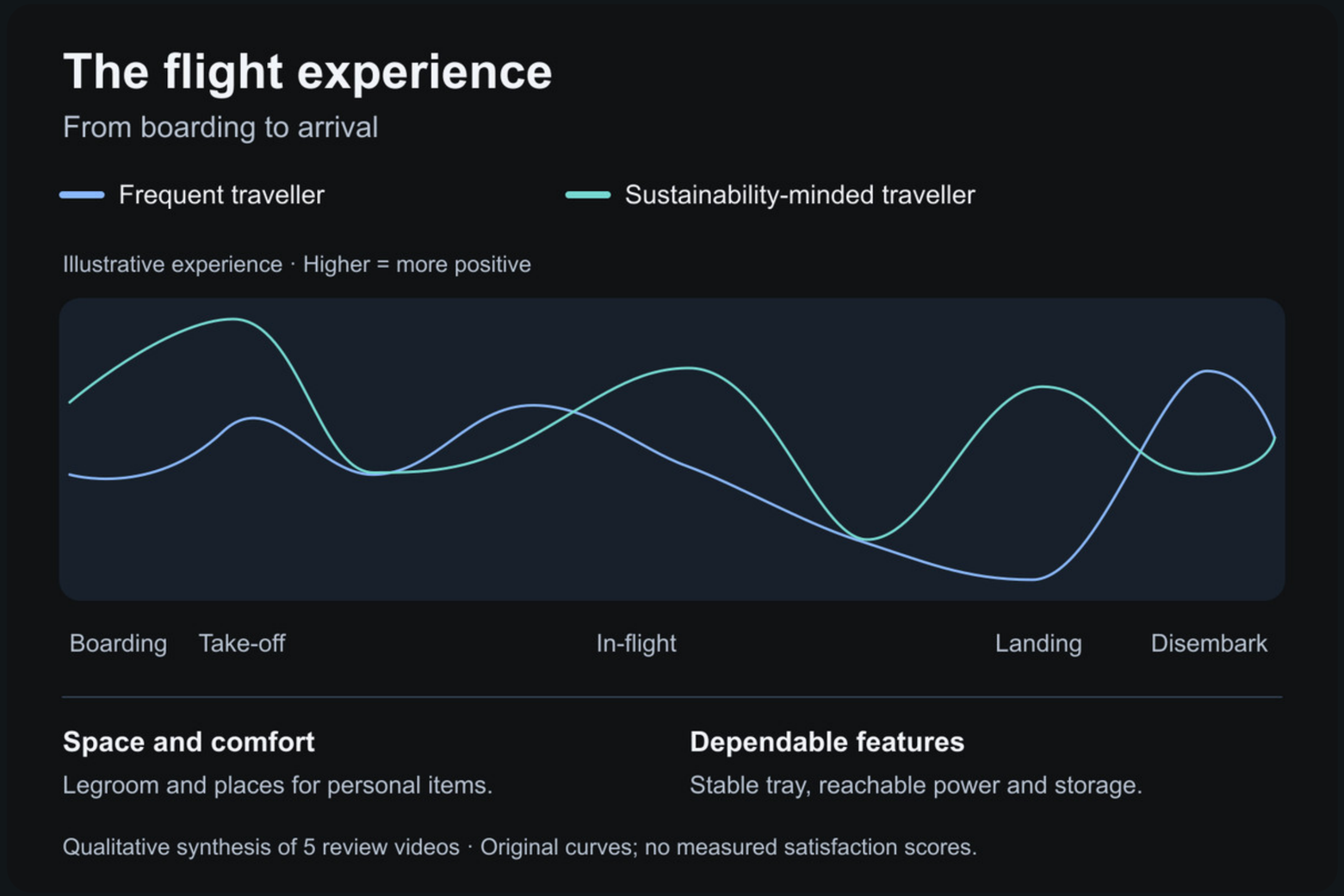


## 89% / 94.7%

Reported improved viewing experience / interaction with music in the simulated evaluation.

No live-street trial. Connected modules, accessibility, outdoor durability and audience retention still need testing.

# Design for the journey.




## Shared research

Team 12 analysed five economy-flight review videos to map two illustrative journeys.

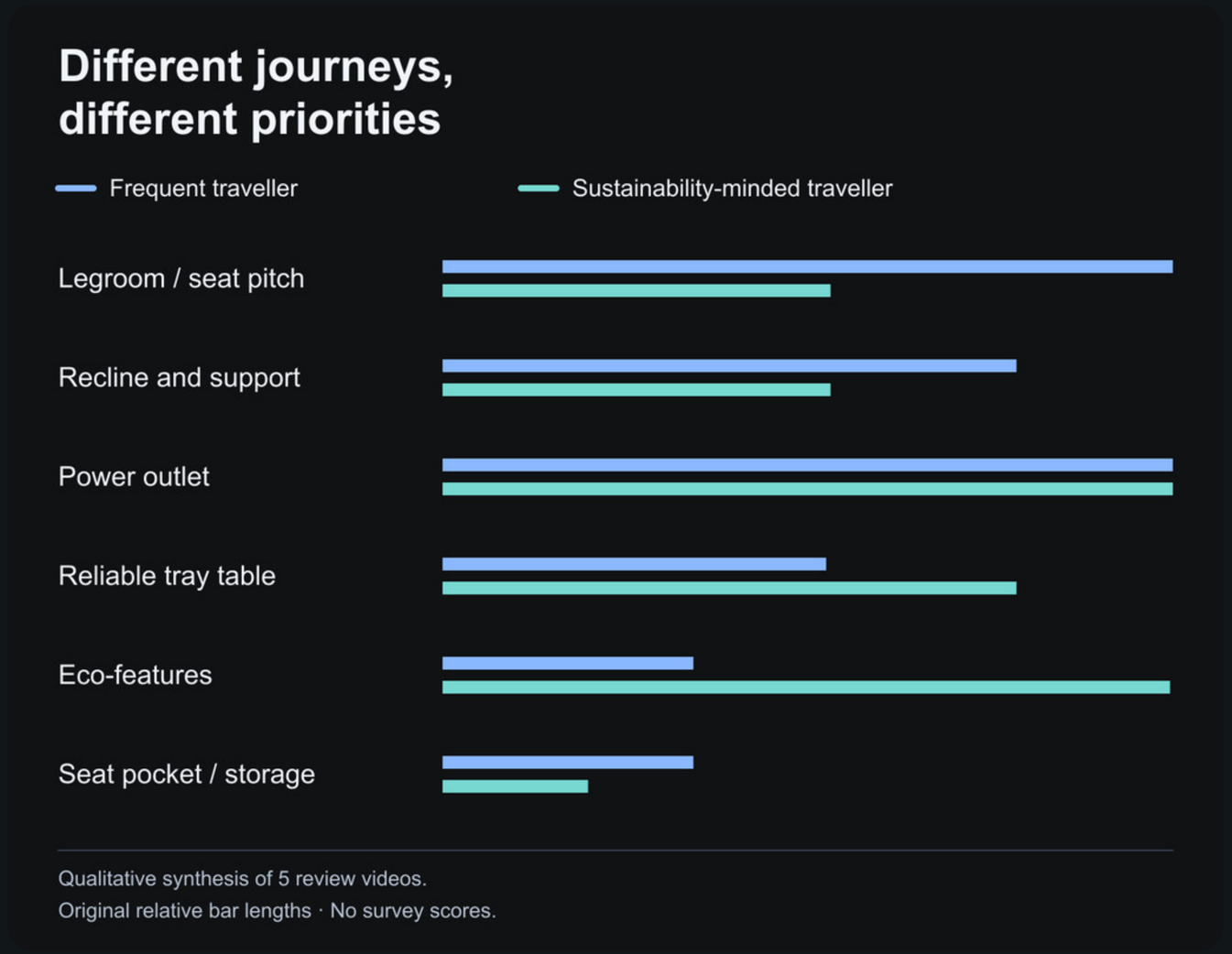
Qualitative secondary research. These curves are not measured satisfaction scores.

## My design response

A shared supporting frame with replaceable cushioning, electronics and accessories, configured around the flight.

I also joined a 40-minute interview with a cabin maintenance engineer.

# A service route for every module.




01 **Choose features**
Select the optional equipment at booking.

02 **Inspect and log**
Staff record damaged or stained modules after the previous flight.

03 **Prepare and fit**
Use the configuration and maintenance record to prepare replacements.

04 **Use during the flight**
The configured seat supports the passenger's activities on board.

05 **Clean, repair or separate**
Treat removed modules off the aircraft according to their condition.

06 **Return to the shared pool**
Serviceable parts become available for another seat.

# Keep the frame. Renew the parts.



AI-refined concept from project CAD, with inferred finishing details. Not a built seat.



Original CAD: repeated aluminium frame members support replaceable cushioning and accessories.

## My contribution

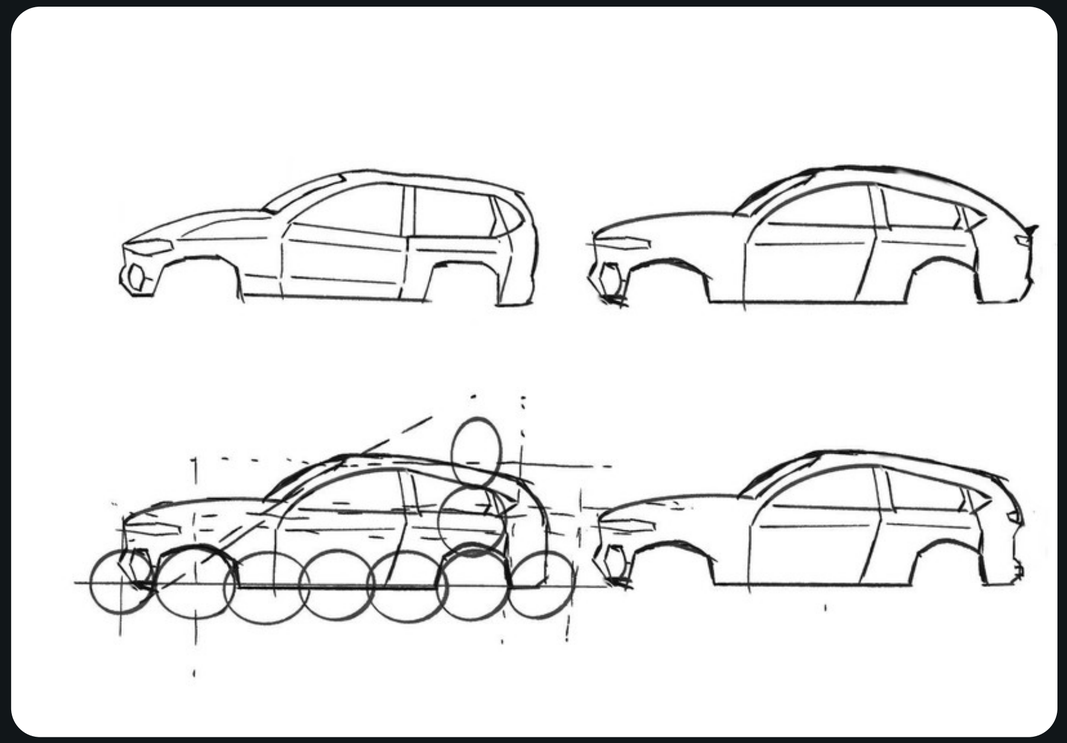
Individual concept, CAD, service system and environmental evaluation; built on Team 12 research.

## Next validation

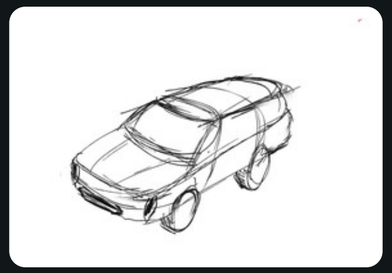
Prototype fit, comfort and repeated connector use; then structural and certification assessment.

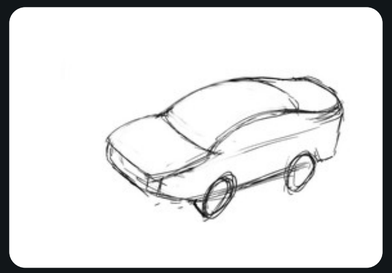
# A practical SUV, shaped by airflow

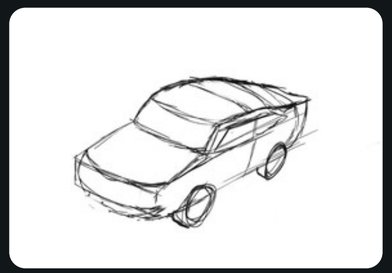
Benchmarking the Volvo EX90, Model Y and Cayenne focused my research on roof-to-tail flow, wheel areas and underbody exposure. The goal was to retain cabin and luggage space while softening the rear transition.



Original proportion studies. The alternatives are qualitative shape choices, without a measured drag or cargo-volume comparison.







## Selected: compact tailgate

A gently sloping roof and short tailgate balance the rear transition with useful luggage space.

## Continuous fastback

A more rounded roof-to-tail transition, with a greater compromise in luggage space.

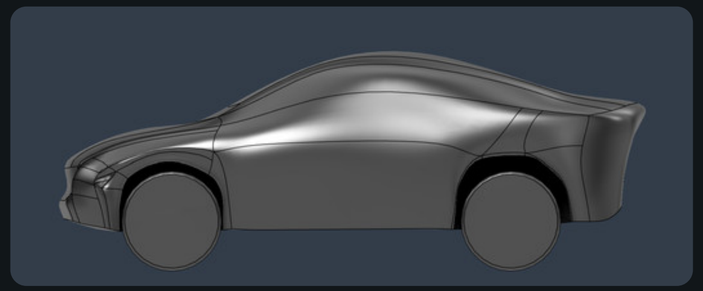
## Upright SUV rear

A flatter roof and vertical tailgate retain the conventional, spacious SUV profile.

# From simplified CAD to the final concept



Final concept visualisation developed from the profile sketches using Vizcom. The CFD used the separate simplified Fusion model.



Simplified, mirrored Fusion geometry used for the aerodynamic study.

## A first aerodynamic check

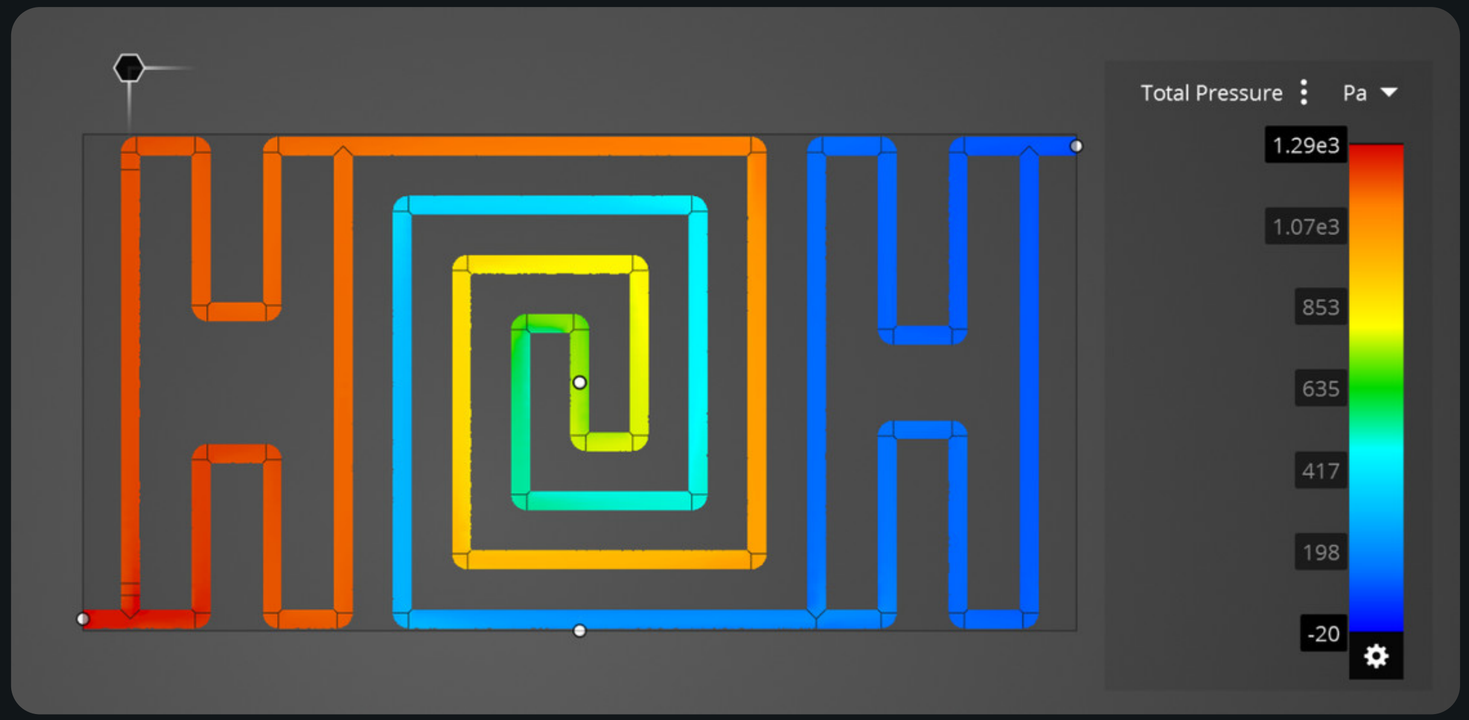
I examined the body in Fluent. The rear wake and wheel areas became priorities for refinement.

The study did not validate the detailed concept or its proposed active features.

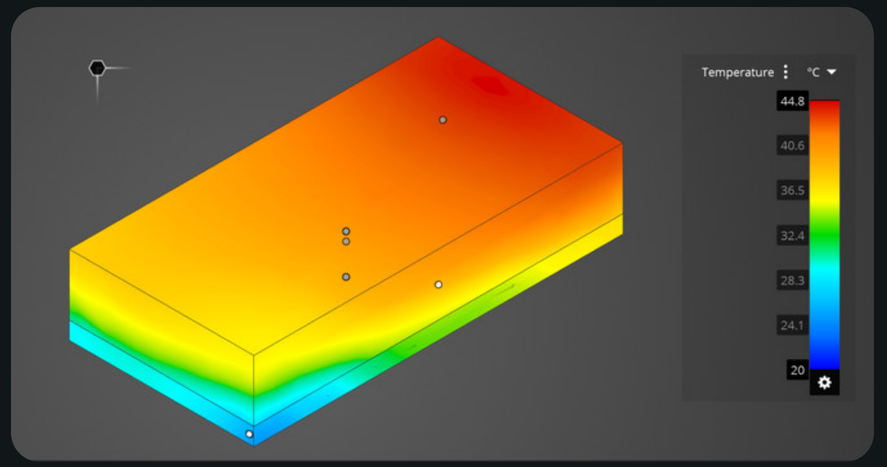
# Battery cooling: reducing resistance

I contributed channel and aspect-ratio research, CAD and design iterations, plus CFD assistance and results interpretation. The cooling design and simulations are team work.

Reported simulation outputs. Mesh independence and matched boundary settings are unconfirmed; no physical validation is claimed.




Fewer turns and a 3.40 m route: 1.26 kPa reported loss, versus 6.1 kPa initially. Profile 1: 8 mm wide × 6.67 mm deep.




Profile 1: 44.8 °C maximum, 38.5 °C average. The hotter outlet-side region remained.

## Next: rebalance cooling

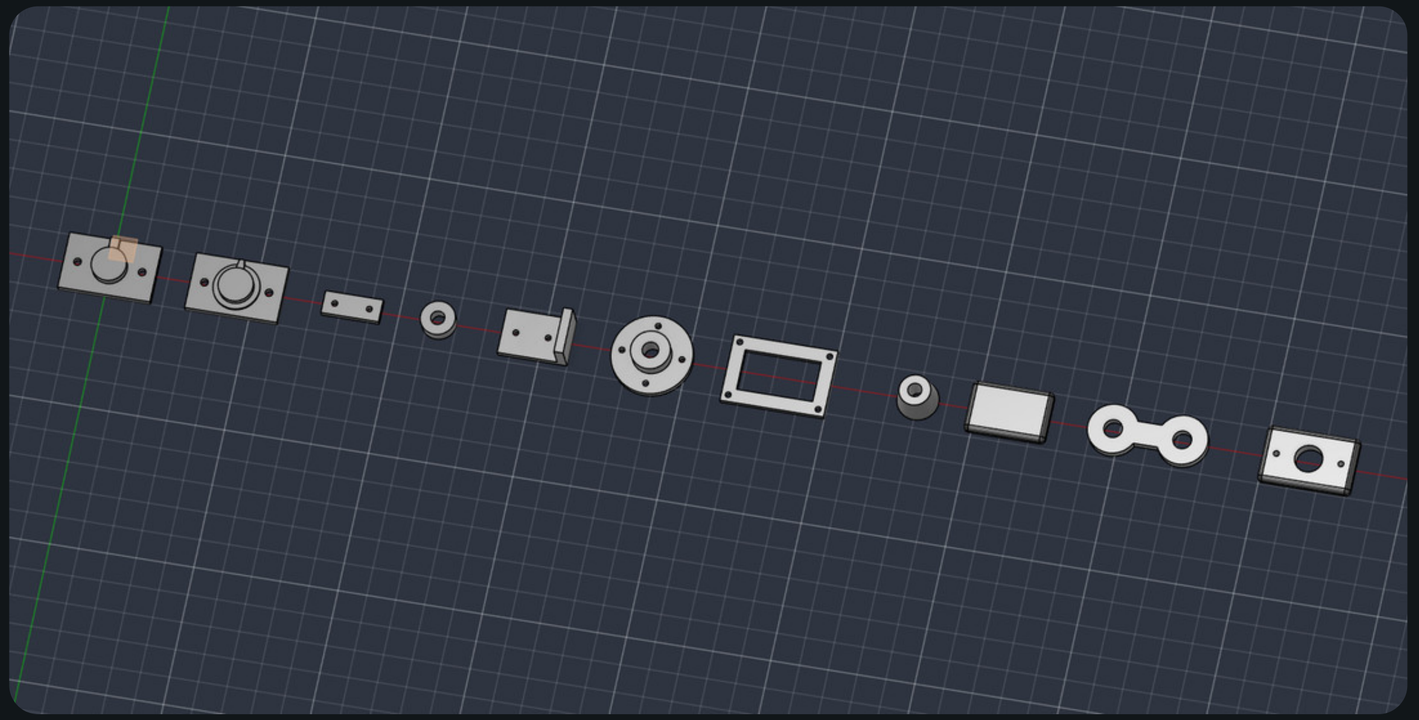
Smaller spiral routes and denser outlet-side coverage were proposed to address uneven cooling. These were not simulated fixes.

# Make design intent checkable.

Humanoid Design
AI-assisted CAD skill

A reusable Codex skill that turns a mechanical brief into native CAD, with design intent and model checks kept visible.

I shaped the engineering questions, review rules and iteration criteria with Codex. This personal tool project is separate from the S00 internship.



Original Fusion trials, 3 Sep 2026 · geometry examples; no physical validation.

## Brief → model → evidence

### 01 / Define
Function, interfaces, units and assumptions.

### 02 / Plan
Check the document and available MCP tools.

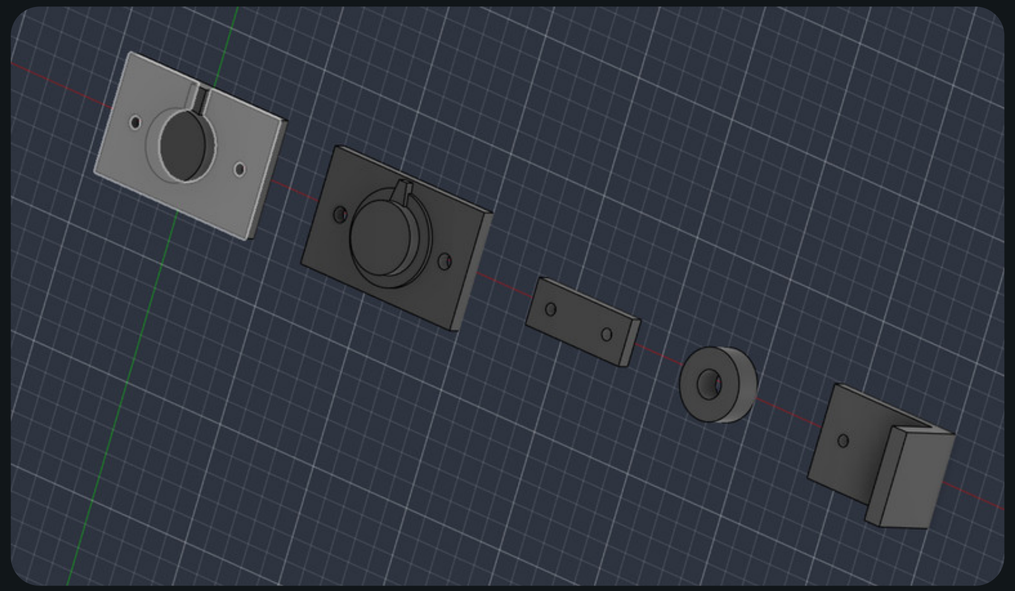
### 03 / Build
Use native features and retain a recovery route.

### 04 / Inspect + revise
Measure geometry, re-identify changed entities and report limits.

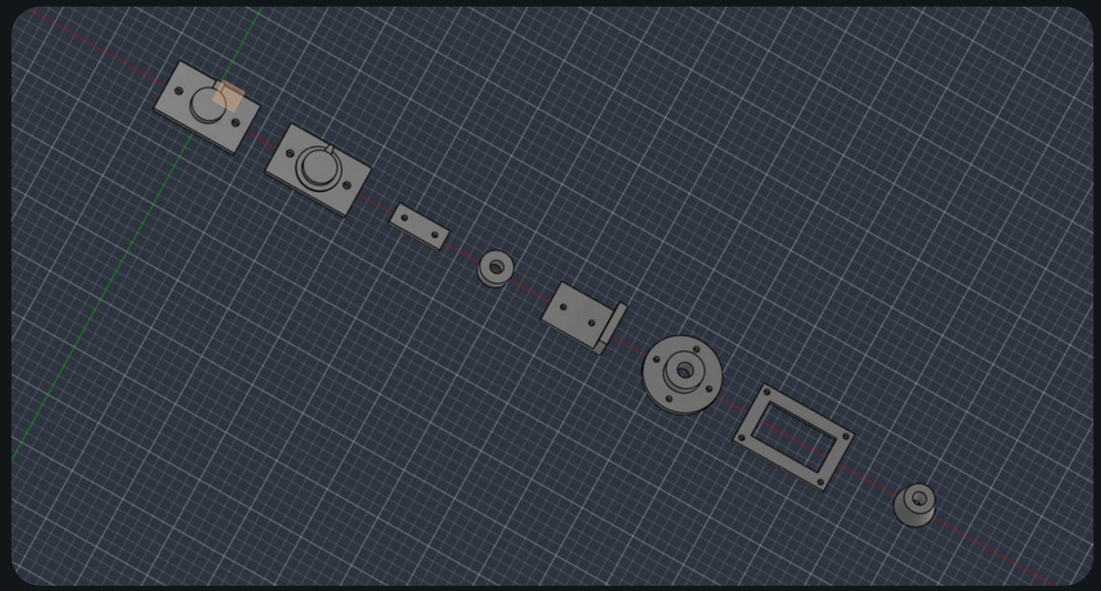
# Native trials shaped the workflow.

## Engineering record → Codex skill → MCP adapters → native Fusion → verification + recovery

Extensions to an existing third-party MCP bridge; no trained CAD model or new kernel. SOLIDWORKS/Ansys were not live-tested in this case.



Clamps, spacer and sensor foot. Tool access and physical fit remained unverified.



Flange, mounting frame and fairlead. A drafted extrusion was the limited fallback for a blocked loft chain.

### Recorded native pass

4 Sep: 19 native checks; 39 static tests.
Scratch geometry discarded.

### Measured model response

Height response: 10 → 18 mm.
Resolved edge: 2 mm fillet.

### In development

Broader briefs, assembly/thread checks and independent reliability evaluation.

# Project Revival: follow the development.

A personal series turning recognisable screen objects into printable components and hands-on builds. Each revival follows its own path from reference to CAD, making and tests.






## Anvil / a first print, unfinished internals

Original eleven-component Fusion architecture and a confirmed full-scale first print in China. Internal and sound development continues.

## Ghost / research and a system plan

A floating desk companion with a lit eye and speech. Component choices remain open; no integrated prototype is demonstrated.

Intended direction: accessible builds for younger makers. Workshops and educational outcomes have not yet been delivered. Images show CAD/identity concepts.

# Anvil: from CAD to a first print.

01

CAD RECORDED

## Translate the reference

My original Fusion architecture: eleven separate components for a nonfunctional prop.

02

FIRST PRINT CONFIRMED

## Make the first iteration

The full-scale first print remains in China. These images show CAD, not that physical print.

03

IN PROGRESS

## Continue the internals

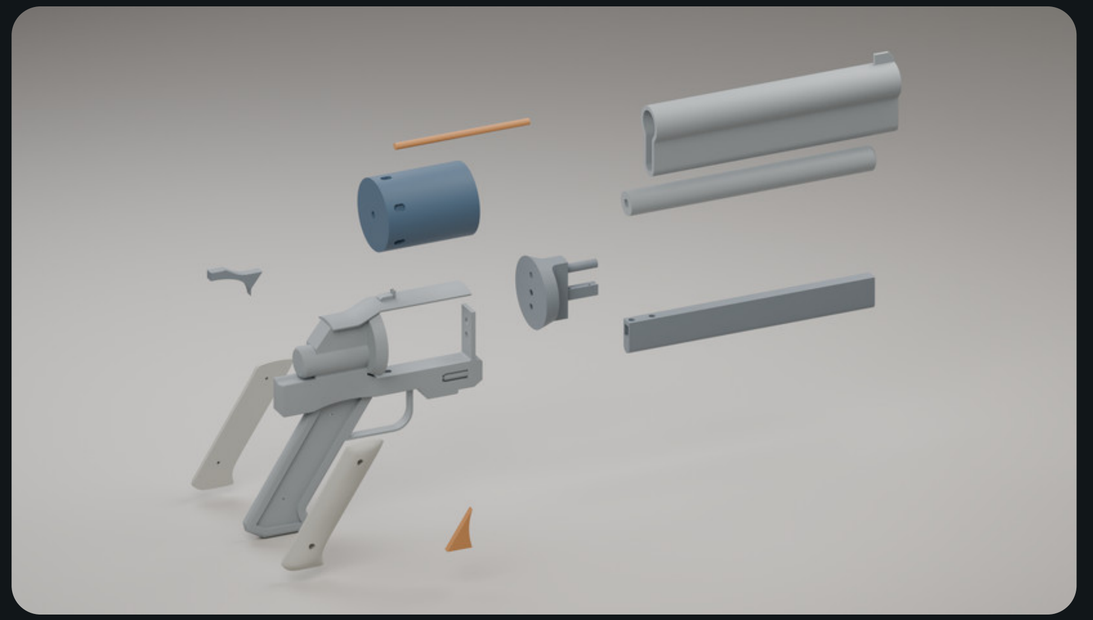
Internals and sound remain unfinished. Later CAD studies do not establish physical operation.

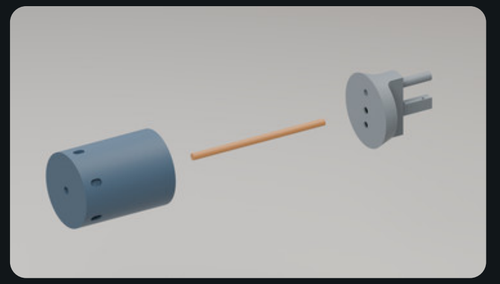
04

NEXT

## Check the assembled prop

Check fit, assembly, movement, durability and sound before calling the internal design complete.





Original CAD and proposed finish visualisations. A first print is confirmed; functioning internals and physical performance are not.

# Ghost: from research to a build plan.

01 RESEARCH RECORDED

## Define the desk companion

The direction: a Destiny-inspired floating shell, a lit eye and spoken conversation.

02 PLAN DOCUMENTED

## Separate the systems

Stock levitation supports the shell; computing and audio stay in a separate stationary pod.

03 IN PROGRESS

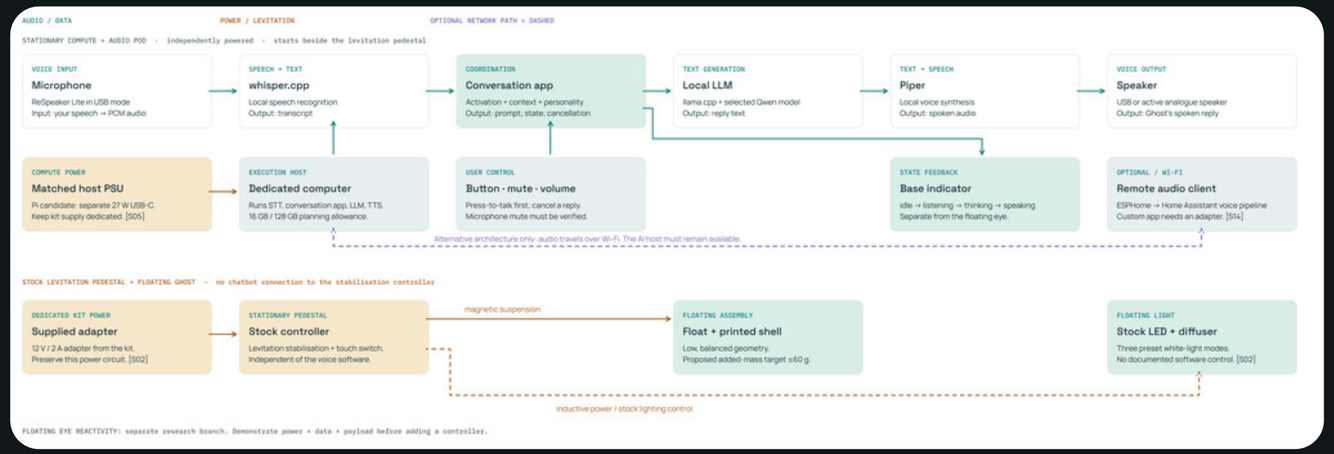
## Resolve the build choices

Computer, enclosure and eye behaviour remain open. Current work is research and planning.

04 NEXT

## Prototype in stages

Test stock levitation and a representative load, prove voice, then print and integrate the shell.



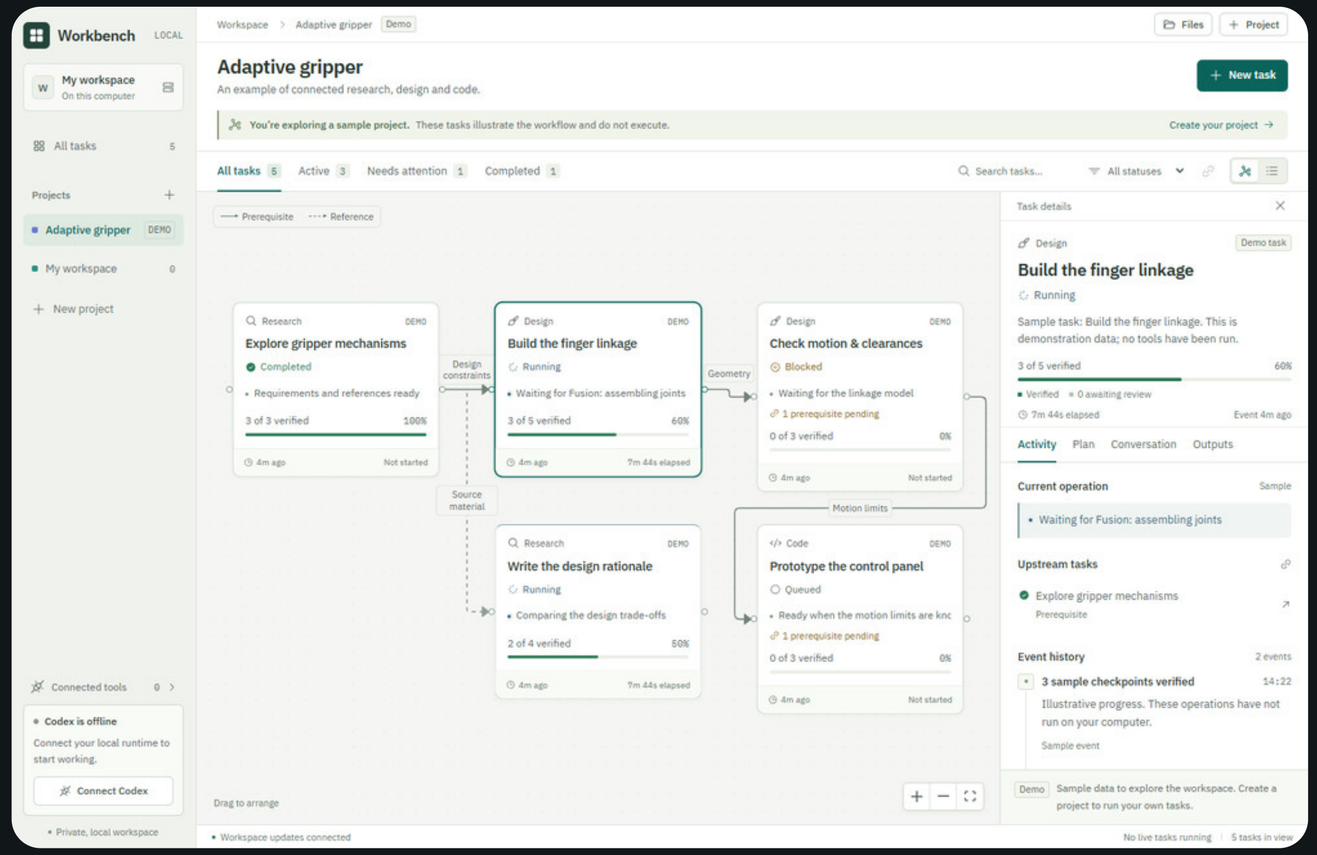

Documented system plan, not an integrated working build. No physical performance or completed voice interaction is claimed.

Boxuan Wang / Imperial Design Engineering

# Keep the work visible as it develops.

Product and workflow design, developed with Codex. A working local prototype; one active turn at a time.

React / TypeScript · Electron · Node.js / SQLite · recorded checks: 19 Oct 2026




Demonstration tasks only; this image does not show live robot or CAD operations.

**01 Workspace · recorded**

Task graph, list, inspector and dependency links; local persistence.

**02 Execution + review · recorded**

Codex execution and checkpoints checked. One active task turn.

**03 Source evidence · recorded**

FTS5 lexical search; exact evidence, revisions and restart checked with fixtures.

**04 Broader use · next**

Real-question retrieval evaluation, deletion controls and native application checks.

**05 Distribution · later**

Installer and updates. Parallel scheduling awaits ownership and recovery checks.


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